ZHAODONG GUO

Academic CV - PhD Applications

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RESEARCH INTERESTS

Efficient & Structured Foundation Models; Long-Context & Functional Memory; State-Space Models & Efficient Attention; Time-Series Foundation Models & Irregular Sequences; Agent Memory & Test-Time Adaptation

RESEARCH PROFILE

MRes student in Artificial Intelligence and Machine Learning at Imperial College London, supervised by Dr. Yingzhen Li. I study compact, reusable memory representations for long and irregular sequences, with current work on coordinate-queryable functional attention (HiPPO-FM) and bounded-memory online spatiotemporal inference (KronHiPPO-STGP). My broader interests are efficient and time-series foundation models, state-space and efficient-attention architectures, and persistent memory for agentic systems.

EDUCATION

Imperial College London - MRes in Artificial Intelligence and Machine Learning *2025 - 2026*

London, UK | Supervisor: Dr. Yingzhen Li

Hebei University of Technology - B.Eng. in Internet of Things Engineering *2021 - 2025*

Tianjin, China | Sino-Foreign Joint Programme | Weighted average: 91.3/100

- Selected grades: Probability and Statistics 96; Linear Algebra 95; Advanced Mathematics 92.

Massey University, Auckland Campus - Bachelor of Information Sciences *2024 - 2025*

Auckland, New Zealand | Joint Bachelor Programme | Weighted average: 87.25/100

SELECTED PUBLICATION

Zhaodong Guo, Naoki Kiyohara, Wenlong Chen, Harrison Bo Hua Zhu, Yingzhen Li. "Analytic Interdomain Memory for Efficient Online HiPPO-SVGP." Accepted poster, ICML 2026 Workshop on Structured Probabilistic Inference & Generative Modeling (SPIGM), Seoul, South Korea, 2026. [ICML 2026 Workshop]

RESEARCH EXPERIENCE

Imperial College London - Research Project *2026 - Present*

HiPPO Functional Memory: Coordinate-Queryable Attention for Irregular Time Series

- Developed HiPPO-FM, a coordinate-queryable functional-memory attention operator that compiles an irregular history into a fixed-rank state with O(Md) persistent storage independent of sequence length, enabling direct readout at arbitrary temporal coordinates without re-encoding the history.
- Designed a matched write/read mechanism using HiPPO-LegS coordinates, interdomain kernel features, and Gram-based decoding; masked self-attention acts before functional compression so the retained state remains content-dependent while preserving a shared coordinate geometry for writing and querying.
- Characterised the induced finite-dimensional functional query space and derived a pointwise approximation bound that separates functional-geometry error from learned-memory mismatch, clarifying what information can and cannot be recovered from the compressed state.
- Established an encode-once, query-many evaluation interface: the same cached functional state can be reused across query orderings, groupings, and resolutions without reprocessing the observation history; numerical checks show coordinate-wise differences below 2.7×10^{-5}.
- Evaluated on Spiral reconstruction, 20 UEA irregular-classification datasets, PhysioNet extrapolation, and marked-event prediction; achieved the best overall UEA rank (1.0), 21.1% lower PhysioNet hidden-region MSE than CRU, and a 22.8× reduction in mean epoch time on the matched PhysioNet benchmark.
- Manuscript under review.

Imperial College London - MRes Research Project *Mar 2026 - Present*

KronHiPPO-STGP: Online Bayesian Spatiotemporal Modeling with Evolving Memory

- Developed KronHiPPO-STGP, a bounded-memory online Bayesian spatiotemporal model that jointly updates a global parametric trend and a Gaussian-process residual through a fixed-size HiPPO interdomain representation, without revisiting past observations.
- Formalised why decoupled online linear-then-GP updates produce stale historical residual targets when the trend is revised, motivating a joint Gaussian state that retains trend-residual cross information across sequential updates.
- Derived joint natural-parameter updates that preserve trend-residual dependence and project accumulated natural parameters between successive HiPPO inducing representations; at M_t=128, projection reduced predictive Gaussian KL from 2.07 to 1.26×10^{-4} and RMSE from 0.2651 to 0.0795.
- Exploited separable Kronecker structure with Schur-complement and generalized Sylvester solves, reducing balanced posterior computation from $O(M^6)$ to $O(M^3)$ time and from $O(M^4)$ to $O(M^2)$ memory; achieved a 1036.7× solver speed-up at M_s=M_t=64 with $<5\times10^{-12}$ relative error versus dense factorization.
- Validated on ERA5-Land, COVID-19, and PEMS-BAY online prediction, reducing RMSE by 66% on ERA5-Land and 5.8% on PEMS-BAY versus the matched fixed-basis variant while maintaining memory cost independent of stream length.
- Manuscript under review.

RESEARCH EXPERIENCE - CONTINUED

Imperial College London - ICML 2026 SPIGM Workshop Research Project *2025 - 2026*

Analytic Interdomain Memory for Efficient Online HiPPO-SVGP

- Context: extended OHSVGP (Chen et al., NeurIPS 2025), which uses time-varying HiPPO projections as interdomain inducing variables to maintain a recurrent compressed memory of streaming observations. [ICML 2026 Workshop]
- Derived closed-form one-dimensional HiPPO-LegS interdomain covariance features by decomposing the temporal factor into sine/cosine components and evaluating the resulting Legendre-weighted integrals through spherical-Bessel identities.
- Developed stable high-order spherical-Bessel evaluation using Miller's backward recurrence, avoiding fragile direct evaluation and enabling accurate analytic feature construction over large polynomial orders.
- Replaced temporal ODE recursion with direct analytic covariance construction, removing integration-step dependence and ODE discretisation error while preserving the recurrent HiPPO interdomain-memory mechanism.
- Validated on Solar Irradiance and COVID-19 mortality streams with comparable predictive accuracy and improved computational scaling; at M=128 and T=3000, analytic kernel construction used 0.236 GFLOPs versus 101.30 GFLOPs for the ODE baseline.

SELECTED PROJECTS

RepoRL: Executable-Reward Post-Training for Repository-Level Coding Agents *2026 - Present*

- Built a repository-level coding agent framework with a bounded ReAct-style AgentRunner and typed tool interfaces for code search, file inspection, patch generation, and targeted test execution; designed isolated Docker-based agent/verifier environments to prevent hidden-test leakage and reward contamination.
- Developed an executable-reward post-training pipeline with sealed SWE-style task manifests, repository-lineage splits, immutable trajectory storage, terminal-grounded rewards, and cost-aware evaluation for coding-agent reinforcement learning.
- Implemented QLoRA supervised fine-tuning and iteration-level on-policy grouped GRPO with fresh current-policy rollouts and policy-identity verification; established reproducible CPU-Docker/GPU orchestration and repository-disjoint evaluation using task success, Pass@k, tool efficiency, invalid-action rate, token usage, and cloud-cost metrics.

MiniMind-LongMem: Memory-Efficient Long-Context LLM Architecture *2026 - Present*

- Built a transparent PyTorch decoder-only Transformer training framework based on MiniMind, implementing BPE tokenization, GQA, RoPE, RMSNorm, SwiGLU, pretraining, and supervised fine-tuning pipelines for controlled architecture research.
- Extended the baseline toward memory-efficient long-context language modeling by integrating local attention with fixed-rank HiPPO functional memory, enabling persistent sequence representations beyond standard attention context windows.
- Designed evaluations comparing memory-augmented architectures with GQA and compressed-KV baselines on 4K-32K context lengths, measuring language-model quality, perplexity, KV-cache memory, peak VRAM consumption, and decoding throughput.

Agricultural Q&A Pair Generation and Dialogue System - Undergraduate Thesis *2024 - 2025*

- Built and curated a 10,000+ pair agricultural-domain question-answering corpus from USDA sources, including cleaning, deduplication, normalisation, and controlled dataset splits.
- Fine-tuned LLaMA 3.1-8B with LoRA and benchmarked it against zero-shot prompting using BLEU, ROUGE, and BERTScore.

AWARDS & INDUSTRY EXPERIENCE

- First Prize, Hebei Division - China Undergraduate Mathematical Contest in Modeling, 2023.
- Honorable Mention - Mathematical Contest in Modeling, COMAP, 2023.

Tianjin Hongmeng Ecological Technology Co., Ltd. - Huawei Cloud Technology Intern *Jan - Feb 2024*

- Supported API deployment, debugging, and cloud-based AI service integration in a production development environment.

RESEARCH & TECHNICAL SKILLS

ML & Sequence Modeling: PyTorch, Hugging Face, HiPPO, state-space models, efficient attention, functional memory, irregular/online time series, spatiotemporal modeling

Probabilistic & Numerical Methods: Gaussian processes, sparse variational inference, interdomain/kernel methods, RKHS approximation, Kronecker algebra, Sylvester equations, Schur complements, low-rank methods

Programming & Research Tools: Python, Linux, Git, LaTeX, runtime/memory profiling; LLM fine-tuning with LoRA and LLaMA 3.1-8B